Chapter 11

Linear Theory for Periodic Orbits

Consider a smooth system of differential equations

$$\dot{x} = f(x), \quad x \in \mathbb{R}^n \tag{11.1}$$

with a periodic solution $\gamma\colon \mathbb{R} \to \mathbb{R}^n$ of period p. The theory of Chapter 10 provides a means of proving the existence of γ. In this chapter we address the question of understanding the dynamics in a neighborhood of the periodic orbit $\Gamma := \gamma([0,p])$. While the philosophy is similar in spirit to that of Chapter 4 – use the dynamics of the linear approximation near the periodic orbit to characterize the local nonlinear dynamics – making use of the linear approximation is much more challenging. The first step, which is discussed in Section 11.1, is to realize that the asymptotic behavior near the periodic orbit can be identified with the behavior of the dynamics generated by a function called the Poincaré map. In particular, the periodic orbit is identified with a fixed point of the Poincaré map and Section 11.2 demonstrates how at the level of eigenvalues and eigenvectors the linearized dynamics near the periodic orbit is related to the linearized dynamics about the fixed point of the Poincaré map. This is expanded upon in Section 11.3 where the trajectories of the linearized flow are encoded via a periodic matrix function.

11.1 Poincaré Sections

Our goal is to adapt the previous concepts of stability of fixed points to periodic orbits and to set up a framework in which stability or lack thereof can be readily determined. Throughout this section let $\varphi\colon \mathbb{R} \times \mathbb{R}^L \to \mathbb{R}^L$ denote the flow associated to (11.1), $\gamma\colon \mathbb{R} \to \mathbb{R}^L$ denote a solution of period p, and $\Gamma := \gamma([0,p])$ denote the associated periodic orbit.

Definition 11.1.1. A periodic solution γ is *stable* if for every open neighborhood $U \subset \mathbb{R}^L$ of Γ, there exists an open set W such that if $x \in W$, then $\varphi(t,x) \in U$ for all $t \geq 0$. A periodic solution γ is *asymptotically stable* if, in addition to being stable, for every $x \in W$,

$\omega(x,\varphi) = \Gamma$. Finally, a periodic solution γ is *asymptotically stable with asymptotic phase* if, in addition to being asymptotically stable, for every $x \in W$ there exists a number $\tau = \tau(x)$ such that $\lim_{t\to\infty} |\varphi(t,x) - \gamma(t-\tau)| = 0$.

At first glance it may appear that verifying the condition of asymptotically stable with asymptotic phase is more difficult than verifying than stable or asymptotically stable, since the latter two involve studying the flow in a bounded neighborhood of Γ whereas the first condition involves a limit as $t \to \infty$. Of course, however, all three conditions are about asymptotic dynamics. What has not been explicitly exploited is that φ is a periodic function. This is the purpose of the Poincaré map, which we now define.

Without loss of generality let $v = \gamma(0) \in \mathbb{R}^L$. Since v is an ordinary point, by the Flow Box Theorem (Exercise 6.3.7), there exists a change of variables $g \colon U \to V$ where $U \subset \mathbb{R}^L$ is an open neighborhood of the origin and V is an open neighborhood of v such that $v = h(0)$ and h maps orbits restricted to U of the flow ψ generated by the differential equation

$$\begin{aligned} \dot{y}_1 &= 1 \\ \dot{y}_2 &= 0 \\ &\vdots \\ \dot{y}_L &= 0 \end{aligned}$$

to orbits of the equation $\dot{x} = f(x)$ restricted to V. Let $S \stackrel{\text{def}}{=} U \cap \left(\{0\} \times \mathbb{R}^{L-1}\right) \subset \mathbb{R}^L$. The associated *Poincaré section* is defined by

$$\Sigma \stackrel{\text{def}}{=} h(S).$$

Observe that $v \in \Sigma$. Since v is a periodic point with period p, $\varphi(p,v) = v$. The continuity of φ implies that there exists a neighborhood $\Sigma' \subset \Sigma$ of v in Σ such that $\varphi(p,\Sigma') \subset V$ and hence $h^{-1}(\varphi(p,\Sigma')) \subset U$. The first return time $T \colon \Sigma' \to \mathbb{R}$ is defined by

$$T(x) \stackrel{\text{def}}{=} \min\left\{t > 0 \mid \varphi(t,x) \in \Sigma\right\}.$$

Observe that there is a continuous function $\rho \colon h^{-1}(\varphi(p,\Sigma')) \to \mathbb{R}$ such that $\psi(\rho(u),u) \in S$ and $\psi([\rho(u),0],u) \subset U$ or $\psi([0,\rho(u)],u) \subset U$, depending on the sign of $\rho(u)$. Therefore, T is continuous and hence $P \colon \Sigma' \to \Sigma$ defined by

$$P(x) \stackrel{\text{def}}{=} \varphi(T(x),x)$$

is continuous. The function P is called a *Poincaré map* for v.

By definition $P(v) = v$. Thus, at least conceptually, the Poincaré map allows us to reduce the problem of stability of γ to understanding stability in the context of the action of P in a neighborhood of v.

With this in mind we formalize the definition of stability in the context of dynamics generated by a smooth function.

Definition 11.1.2. Let $g\colon \mathbb{R}^L \to \mathbb{R}^L$. A point $u \in \mathbb{R}^L$ is a *periodic point* if there exists $m \geq 1$ such that $g^m(u) = u$. The minimal positive k for which $g^k(u) = u$ is the *period* k of u. A periodic point u is *stable* if for all $\epsilon > 0$ there exists a $\delta > 0$, such that if $\|x - u\| < \delta$, then $\|g^n(x) - g^n(u)\| < \epsilon$ for all $n \in \mathbb{N}$. A periodic point that is not stable is *unstable*. The periodic point is *asymptotically stable* or *attracting* if it is stable and in addition if $\|x - u\| < \delta$, then

$$\lim_{n\to\infty} \|g^n(x) - g^n(u)\| = 0.$$

Observe that if $u \in \Sigma$ is a periodic point with period k of the Poincaré map P, then $u \in \mathbb{R}^L$ lies on a periodic orbit of period kp for (11.1).

As in the case of equilibria for flows, the dynamics near a fixed point or periodic point u of a map g can be characterized by the linear dynamics.

Theorem 11.1.3. *Let u be a periodic point for $g\colon \mathbb{R}^L \to \mathbb{R}^L$ with period k. Assume all eigenvalues λ of $Dg^k(u)$ satisfy $|\lambda| < \mu$ for some $\mu \in (0,1)$.*

1. *There exists a norm $\|\cdot\|_*$ on $\mathbb{R}^L$ and a neighborhood $U \subset \mathbb{R}^L$ of u such that for any $x \in U$,*
$$\|g^{jk}(x) - g^{jk}(u)\|_* \leq \mu^j \|x - u\|_* \quad \textit{for all } j \geq 0.$$
2. *For any norm $\|\cdot\|$ on $\mathbb{R}^L$ there exists a neighborhood $U \subset \mathbb{R}^L$ of u and a constant $C \geq 1$ such that for any $x \in U$,*
$$\|g^{jk}(x) - g^{jk}(u)\| \leq C\mu^j \|x - u\| \quad \textit{for all } j \geq 0.$$

The proof is similar to that of Theorem 4.3.3. The following corollary is an important consequence.

Corollary 11.1.4. *Let u be a periodic point for $g\colon \mathbb{R}^L \to \mathbb{R}^L$ with period k. If all eigenvalues λ of $Dg^k(u)$ satisfy $|\lambda| < 1$, then u is asymptotically stable.*

The following definition is analogous to Definition 4.3.4.

Definition 11.1.5. Let $A\colon \mathbb{R}^L \to \mathbb{R}^L$ be a linear map. The *stable*, *unstable*, and *center eigenspaces* are

$$\mathbb{E}^s_A \stackrel{\text{def}}{=} \text{span}\,\{v \mid v \text{ is a generalized eigenvector of an eigenvalue } \lambda \text{ of } A \text{ with } |\lambda| > 1\}$$
$$\mathbb{E}^u_A \stackrel{\text{def}}{=} \text{span}\,\{v \mid v \text{ is a generalized eigenvector of an eigenvalue } \lambda \text{ of } A \text{ with } |\lambda| < 1\}$$
$$\mathbb{E}^c_A \stackrel{\text{def}}{=} \text{span}\,\{v \mid v \text{ is a generalized eigenvector of an eigenvalue } \lambda \text{ of } A \text{ with } |\lambda| = 1\}$$

respectively.

Definition 11.1.6. A periodic point u for a smooth map $g\colon \mathbb{R}^L \to \mathbb{R}^L$ with period k is *hyperbolic* if given $\lambda \in \sigma(Dg^k(u))$, $|\lambda| \neq 1$.

Observe that for a hyperbolic periodic point u the center eigenspace is $\{\mathbf{0}\}$.

In Section 4.4 we state the Hartman-Grobman theorem for flows that relates the behavior of a nonlinear flow near a hyperbolic fixed point to the appropriate linear dynamics. There is a corresponding linear theory for fixed points of maps.

Theorem 11.1.7 (Hartman-Grobman Theorem). *Let $g\colon \mathbb{R}^L \to \mathbb{R}^L$ be a diffeomorphism with hyperbolic fixed point p. Then, there exists neighborhoods V of p and U of $\mathbf{0}$ and a homeomorphism $h\colon U \to V$ such that*

$$g(h(x)) = h(Dg(p)x)$$

for all $x \in U$.

For a proof see [34, Theorem 5.6.2].

To put the content of this section into context recall that our goal is to understand the behavior of solutions to (11.1) near the periodic solution γ. The Poincaré map $P\colon \Sigma' \to \Sigma$ has a fixed point $v = \gamma(0)$. Theorem 11.1.7 implies that if $DP(v)$ is hyperbolic, then the generalized eigenvalues and eigenvectors of $DP(v)$ inform us about the asymptotic dynamics of P in a neighborhood of v, and therefore, we understand the asymptotic dynamics of solutions in a neighborhood of Γ. In fact (see Theorem **??**) this is sufficient information for us to understand asymptotic stability with asymptotic phase.

Of course in practice we are given (11.1), not the Poincaré map. Thus we need to be able to determine the eigenvalues of $DP(v)$ from (11.1). This is discussed in Section 11.2.

11.2 Linear Stability of Periodic Orbits

We now develop analytic techniques for determining the stability of the periodic orbit based on the *variational equation* about the periodic solution, i.e.,

$$\dot{u} = Df(\gamma(t))u. \tag{11.2}$$

To motivate how the variational equation comes into play observe that we are interested in studying the behavior of solutions $x(t)$ that are close to $\gamma(t)$ and thus it is convenient to use the coordinates $u(t) = x(t) - \gamma(t)$ where it is assumed that $\|u\| \ll 1$. Using the Taylor expansion of f around γ we obtain

$$\begin{aligned}
\dot{x} &= f(x)\\
\dot{u} + \dot{\gamma} &= f(\gamma + u)\\
\dot{u} + \dot{\gamma} &= f(\gamma) + Df(\gamma(t))u + \text{h.o.t}\\
\dot{u} &= Df(\gamma(t))u
\end{aligned}$$

where the last line is obtained by ignoring the higher order terms and using the fact that γ is a solution to the differential equation, i.e., $\dot{\gamma} = f(\gamma)$.

Observe that the variational equation (11.2) is linear and falls in the more general category of homogeneous linear systems of the form

$$\dot{x} = A(t)x \tag{11.3}$$

where $A\colon \mathbb{R} \to M_L(\mathbb{R})$ is a continuous matrix valued function.

A naive guess may lead the reader to believe that stability of the periodic orbit can be obtained by studying the eigenvalues of $Df(\gamma(t))$. The following example indicates that this approach will not succeed.

Example 11.2.1. (Marcus-Yamabe) *Consider the π-periodic system $\dot{u} = A(t)u$ where*

$$A(t) = \begin{bmatrix} -1 + \frac{3}{2}\cos^2 t & 1 - \frac{3}{2}\sin t \cos t \\ -1 - \frac{3}{2}\sin t\cos t & -1 + \frac{3}{2}\sin^2 t \end{bmatrix}.$$

The eigenvalues of $A(t)$ are $\frac{1}{4}(-1 \pm \sqrt{7}i)$ independent of t. Observe that the real parts of the eigenvalues are negative. However,

$$u(t) = e^{\frac{t}{2}} \begin{bmatrix} -\cos t \\ \sin t \end{bmatrix}$$

is a solution and hence the zero solution is unstable.

As the following result demonstrates, solutions to homogeneous linear systems are defined over $\mathbb{R}$.

Proposition 11.2.2. *Given the homogeneous linear system* (11.3) *the solution to the initial value problem $x(t_0) = x_0$ is defined for all $t \in \mathbb{R}$.*

Proof. The proof is by contradiction. The existence of a solution $\varphi\colon J \to \mathbb{R}^n$ follows from Theorem 6.1.9. Assume that $J = (\alpha, \beta)$, is the maximal interval of existence and $\beta < \infty$. By Lemma 6.1.1

$$\varphi(t) = \varphi(t_0) + \int_{t_0}^{t} A(s)\varphi(s)\, ds. \tag{11.4}$$

By assumption A is continuous over $\mathbb{R}$, hence there exists M such that

$$\sup_{t\in[t_0,\beta]} \|A(t)\| \leq M.$$

Applying this bound to (11.4) we obtain

$$\|\varphi(t)\| \leq \|x_0\| + \int_{t_0}^{t} M\|\varphi(s)\|\, ds,$$

which, by Gronwall's inequality (Theorem 4.4.5), implies

$$\|\varphi(t)\| \leq \|x_0\| e^{M(t-t_0)}.$$

In particular, $\varphi(t)$ is bounded on $[t_0, \beta]$. This contradicts Theorem 6.1.17(b). Therefore $\beta = \infty$. A similar argument shows that $\alpha = -\infty$. $\square$

By as result similar to that of Corollary **??** solutions to the variational equation can be represented using a fundamental matrix solution $\Phi(t)$, i.e., a time dependent invertible matrix with the property that each column is a solution to (11.2). Recall that a fundamental matrix solution $\Phi(t)$ such that $\Phi(\theta) = I$ is the principle fundamental matrix solution at θ.

For any $\theta \in [0, \tau]$, define

$$\gamma_\theta(t) = \gamma(t + \theta)$$

the phase-shift re-parametrization of Γ. Being autonomous, system (11.1) has the property that any of the curves $\gamma_\theta(t)$ is a τ-periodic solution satisfying $\gamma_\theta(0) = \gamma(\theta)$.

Definition 11.2.3. Let $\gamma : \mathbb{R} \to \mathbb{R}^L$ be a τ-periodic solution of (11.1) and let $\Phi_\theta(t)$ be the unique solution of the non-autonomous linear problem

$$\begin{cases} \dot{\Phi}_\theta = Df(\gamma_\theta(t))\Phi_\theta \\ \Phi_\theta(0) = I. \end{cases} \tag{11.5}$$

The matrix $\Phi_\theta(\tau)$ is called the *monodromy matrix* of $\gamma_\theta(t)$.

As we demonstrate below the monodromy matrices $\Phi_\theta(\tau)$ contain the information necessary to determine the stability of Γ. The first step in showing this is to clarify the relationships between the monodromy matrices. The next two results are direct consequences of $\Phi_\theta(t)$ being a fundamental matrix solution. Having chosen $\gamma(t) = \gamma_0(t)$, we simplify the notation by setting $\Phi(\tau) = \Phi_0(\tau)$.

Proposition 11.2.4. *For any $\theta \in [0, \tau]$, the solution $\Phi_\theta(t)$ of (11.5) satisfies*

$$\Phi_\theta(n\tau + t) = \Phi_\theta(t)\Phi_\theta(\tau)^n, \ \forall t \in \mathbb{R}, \ \forall n \in \mathbb{N}.$$

Proof. Without loss of generality, let us consider $\theta = 0$. Denote $A(t) = Df(\gamma(t))$, and therefore $\dot{\Phi}(t) = A(t)\Phi(t)$. We proceed by induction on $n \geq 0$. For $n = 0$ the result is obvious. Suppose the result holds for $n - 1$. Then

$$\Phi(n\tau) = \Phi((n-1)\tau + \tau) = \Phi(\tau)\Phi(\tau)^{n-1} = \Phi(\tau)^n.$$

Let

$$\Psi(t) \stackrel{\text{def}}{=} \Phi(t + n\tau)\Phi(n\tau)^{-1}.$$

It follows that $\Psi(0) = I$ and that

$$\dot{\Psi}(t) = \dot{\Phi}(n\tau + t)\Phi(n\tau)^{-1} = A(n\tau + t)\Phi(n\tau + t)\Phi(n\tau)^{-1} = A(t)\Psi(t).$$

From unicity of solutions of the initial value problem, one has that $\Psi(t) = \Phi(t)$, which implies that

$$\Phi(t) = \Psi(t) = \Phi(t + n\tau)\Phi(n\tau)^{-1} \Longrightarrow \Phi(t + n\tau) = \Phi(t)\Phi(n\tau).$$

Hence,

$$\Phi(t + n\tau) = \Phi(t)\Phi(n\tau) = \Phi(t)\Phi(\tau)^n, \ \forall t \in \mathbb{R}. \quad \square$$

Proposition 11.2.5. *The matrices $\Phi_\theta(\tau)$ and $\Phi(\tau)$ are equivalent under conjugation. In particular*

$$\Phi_\theta(\tau) = \Phi(\theta)\Phi(\tau)\Phi(\theta)^{-1}. \tag{11.6}$$

Proof. The matrix $\tilde{\Phi}(t) \stackrel{\text{def}}{=} \Phi(t+\theta)$ is a solution of the equation $\dot{u} = Df(\gamma(t+\theta))u = Df(\gamma_\theta(t))u$, with $\tilde{\Phi}(0) = \Phi(\theta)$. Since $\Phi_\theta(t)$ is the principal fundamental solution,

$$\tilde{\Phi}(t) = \Phi_\theta(t)\Phi(\theta).$$

It follows

$$\Phi_\theta(t) = \tilde{\Phi}(t)\Phi(\theta)^{-1} = \Phi(t+\theta)\Phi(\theta)^{-1}, \quad \forall t. \tag{11.7}$$

Using Proposition 11.2.4, one can then conclude (with $n = 1$) that

$$\Phi_\theta(\tau) = \Phi(\tau+\theta)\Phi(\theta)^{-1} = \Phi(\theta)\Phi(\tau)\Phi(\theta)^{-1}. \quad \square$$

Proposition 11.2.5 implies that all monodromy matrices $\Phi_\theta(\tau)$ have the same eigenvalues.

Definition 11.2.6. The *Floquet multipliers* of the periodic orbit Γ are the eigenvalues of the monodromy matrix $\Phi(\tau)$.

Proposition 11.2.7. *For any periodic orbit* Γ *of* (11.1) *there is at least one Floquet multiplier with value* 1*.*

Proof. Let $u(t) = f(\gamma(t))$. Then

$$\dot{u} = Df(\gamma(t))\dot{\gamma} = Df(\gamma(t))f(\gamma) = Df(\gamma(t))u.$$

Thus, $t \to f(\gamma(t))$ is a solution to the variational equation with initial condition $f(\gamma(0))$. Hence, since $\Phi(t)$ is the principal fundamental matrix solution of $\dot{u} = Df(\gamma(t))u$, there exists a constant vector $v \in \mathbb{R}^n$ such that $f(\gamma(t)) = \Phi(t)v$, where $v = Iv = \Phi(0)v = f(\gamma(0))$. In addition, because γ is a τ-periodic solution

$$\Phi(\tau)v = f(\gamma(\tau)) = f(\gamma(0)) = v$$

and thus $v = f(\gamma(0))$ is an eigenvector of $\Phi(\tau)$ with eigenvalue 1. $\square$

The Floquet multiplier associated with $v = f(\gamma(0))$ is called the *trivial* Floquet multiplier. We now show that the remainder of the Floquet multipliers are directly related to the linearization of the Poincaré map associated with the periodic orbit. Recall from Section 11.1 that given the periodic solution γ, there is a Poincaré section Σ and an associated Poincaré map $P\colon N \to \Sigma$ for some neighborhood $N \subset \Sigma$ of $p_0 = \gamma(0) \in \Gamma$ such that $P(p_0) = p_0$. Furthermore, the first return time is a differentiable map $T\colon N \to \mathbb{R}$ such that $T(p_0) = \tau$ and $\varphi(T(a), a) \in \Sigma$ for all $a \in N$ where φ is the flow associated with (11.1).

Consider a vector $v \in \mathcal{T}_{p_0}\Sigma$, where $\mathcal{T}_{p_0}\Sigma$ denotes the tangent space of Σ at p_0. Understanding the contraction or expansion of this vector allows us to understand the stability or instability of the fixed point p_0 of the Poincaré map.

Lemma 11.2.8. *$D_x\varphi(T(p_0), p_0) = \Phi(\tau)$.*

Proof. Observe that

$$\begin{aligned}
\frac{d}{dt}\varphi(t,x) &= f(\varphi(t,x)) \\
D_x\frac{d}{dt}\varphi(t,x) &= D_x[f(\varphi(t,x))] \\
\frac{d}{dt}D_x\varphi(t,x) &= Df(\varphi(t,x))D_x\varphi(t,x) \\
\frac{d}{dt}D_x\varphi(t,p_0) &= Df(\gamma(t))D_x\varphi(t,p_0).
\end{aligned}$$

Hence, $\Psi(t) \stackrel{\text{def}}{=} D_x\varphi(t,p_0)$ satisfies $\dot{\Psi}(t) = Df(\gamma(t))\Psi(t)$, $\Psi(0) = I$. From the uniqueness of solutions to the IVP one gets that $\Psi(t) = D_x\varphi(t,p_0) = \Phi(t)$ and therefore $\Phi(\tau) = D_x\varphi(\tau,p_0) = D_x\varphi(T(p_0),p_0)$. □

Thus we want to understand the derivative of the Poincaré map in the direction of v; that is, we want to differentiate

$$P(p_0 + \lambda v) = \varphi(T(p_0 + \lambda v), p_0 + \lambda v)$$

with respect to λ at $\lambda = 0$. This gives

$$\begin{aligned}
DP(p_0)v &= DP(x)|_{x=p_0}v \\
&= \frac{d\varphi}{dt}(T(x),x)|_{x=p_0}\nabla T(x)|_{x=p_0}v + D_x\varphi(T(x),x)|_{x=p_0}v \\
&= \frac{d\varphi}{dt}(T(p_0),p_0)\nabla T(p_0)v + D_x\varphi(T(p_0),p_0)v \\
&= (\nabla T(p_0)v)f(p_0) + \Phi(\tau)v
\end{aligned}$$

where the last line follows from Lemma 11.2.8.

Proposition 11.2.9. *Let γ be a periodic solution of period τ with $\gamma(0) = p_0$. Let P be the Poincaré map associated with γ at p_0. Let $\{\sigma_l \mid l = 1, \ldots, L-1\}$ be the eigenvalues of $DP(p_0)$. Then, the Floquet multipliers of γ are $\{\sigma_l \mid l = 1, \ldots, L-1\} \cup \{1\}$.*

Proof. Let Σ be the Poincaré section associated with P. Since $f(p_0)\cdot v = 0$ for any $v \in \mathcal{T}_{p_0}\Sigma$ we can choose a basis of $\mathbb{R}^n = \langle f(p_0)\rangle \oplus \mathcal{T}_{p_0}\Sigma$ of the form

$$f(p_0), v_1, \ldots, v_{L-1}$$

where $v_l \in \mathcal{T}_{p_0}\Sigma$. Note that in this basis given $v \in \mathcal{T}_{p_0}\Sigma$ we can write

$$v = \begin{bmatrix} 0 \\ v_\Sigma \end{bmatrix} \in \mathbb{R}^L,$$

for some vector $v_\Sigma \in \mathbb{R}^{L-1}$. Using this basis and the fact that $\Phi(\tau)f(p_0) = f(p_0)$, one can represent the monodromy matrix as

$$\Phi(\tau) = \begin{bmatrix} 1 & U \\ 0 & V \end{bmatrix} \tag{11.8}$$

where U and V are $1 \times (L-1)$ and $(L-1) \times (L-1)$ matrices. Given $v \in \mathcal{T}_{p_0}\Sigma$ we can apply (**??**) and the change of basis to conclude that

$$DP(p_0)v = \begin{bmatrix} \nabla T(p_0)v \\ 0 \end{bmatrix} + \begin{bmatrix} 1 & U \\ 0 & V \end{bmatrix} \begin{bmatrix} 0 \\ v_\Sigma \end{bmatrix} = \begin{bmatrix} \nabla T(p_0)v + Uv_\Sigma \\ Vv_\Sigma \end{bmatrix} = \begin{bmatrix} 0 \\ Vv_\Sigma \end{bmatrix},$$

since the range of $DP(p_0)$ is tangent to Σ. Hence, the eigenvalues of $DP(p_0)$ and V are the same. From (11.8), we can conclude that the Floquet multipliers of γ are given by the eigenvalues of V and 1, that is by $\{\sigma_l \mid l = 1, \ldots, L-1\} \cup \{1\}$. □

Proposition 11.2.9 allows us to define hyperbolicity of periodic orbits of ODEs in a manner consistent with that of the associated fixed point for the Poincaré map.

Definition 11.2.10. A periodic orbit for an ODE is *hyperbolic* if its nontrivial Floquet multipliers $\{\sigma_l \mid l = 1, \ldots, L-1\}$ satisfy $|\sigma_l| \neq 1$.

We can now use the Hartman-Grobman theorem for maps to obtain the following stability result for periodic orbits of ODEs.

Proposition 11.2.11. *Let $\Gamma = \{\gamma(t), t \in [0, \tau]\}$ be a τ-periodic orbit of the system* (11.1) *and let $\{\sigma_l\}_{l=1,\ldots,L-1}$ be the corresponding set of nontrivial Floquet multipliers. Then,*

- *Γ is asymptotically stable if for all $l \in \{1, \ldots, L-1\}$, $|\sigma_l| < 1$;*
- *Γ is unstable if there exists $l \in \{1, \ldots, L-1\}$ such that $|\sigma_l| > 1$.*

Moreover, if $p \leq L-1$ Floquet multipliers have modulus less than one, and $q < L-p$ Floquet multipliers have modulus greater than one, Γ is said to have p stable directions and q unstable directions.

While Proposition 11.2.11 attains our goal of an analytic determination of stability of periodic orbits it does not provide a clear description of the behavior of solutions to the differential equation in a neighborhood of Γ. To obtain this we turn to Floquet theory.

11.3 Floquet's Theorem

Floquet's Theorem introduces a canonical decomposition of the principle fundamental matrix solutions of (11.5) in the form of a product of a periodic matrix function and an exponential matrix function. The periodic matrix function provides information about the property of solutions as a function of the phase of the periodic orbit, and the exponential matrix function provides information about the expansion or contraction of the dynamics near the periodic orbit.

We begin with the following theorem from linear algebra [10, Theorem 2.82].

Theorem 11.3.1. *If C is a nonsingular $n \times n$ matrix, then there is an $n \times n$ matrix B (possibly complex) such that $e^B = C$. If C is a nonsingular real $n \times n$ matrix, then there is a real $n \times n$ matrix R such that $e^R = C^2$.*

Theorem 11.3.2. [Floquet's Theorem] *Consider the τ periodic differential equation*

$$\dot{u} = A(t)u. \tag{11.9}$$

Let $\Phi(t)$ be a fundamental matrix solution of (11.9). Then $\Phi(t+\tau)$ is also a fundamental matrix solution and $\Phi(t+\tau) = \Phi(t)\Phi^{-1}(0)\Phi(\tau)$. Also, there exist a constant matrix B (possibly complex) and a τ-periodic matrix function $P(t)$ such that

$$\Phi(t) = P(t)e^{Bt}. \tag{11.10}$$

Moreover, there exist a real constant matrix R and a real 2τ-periodic matrix function $Q(t)$ such that

$$\Phi(t) = Q(t)e^{Rt}. \tag{11.11}$$

Definition 11.3.3. The decomposition (11.10) is called a *complex Floquet normal form* for the fundamental matrix solution $\Phi(t)$ while the decomposition (11.11) is called a *real Floquet normal form* for $\Phi(t)$.

Proof. Define $\Psi(t) \stackrel{\text{def}}{=} \Phi(t+\tau)$. Observe that $\Psi(t)$ is a matrix solution, since

$$\dot{\Psi}(t) = \dot{\Phi}(t+\tau) = A(t+\tau)\Phi(t+\tau) = A(t)\Psi(t).$$

Let $C \stackrel{\text{def}}{=} \Phi^{-1}(0)\Phi(\tau) = \Phi^{-1}(0)\Psi(0)$. This is a non singular matrix. The matrix function $\Phi(t)C$ is a matrix solution of (11.9) with initial condition $\Phi(0)C = \Psi(0)$, since each of its column is a linear combination of the columns of $\Phi(t)$. By uniqueness of the solutions of the initial value problem associated to (11.9), one gets that $\Psi(t) = \Phi(t)C$ for all $t \in \mathbb{R}$. Moreover,

$$\begin{aligned}
\Phi(t+\tau) &= \Phi(t)C = \Phi(t)\Phi^{-1}(0)\Phi(\tau) \\
\Phi(t+2\tau) &= \Phi((t+\tau)+\tau)C = \Phi(t+\tau)C = \Phi(t)C^2.
\end{aligned}$$

By Theorem 11.3.1, there is a matrix $\tilde{B}$ (possibly complex) such that $e^{\tilde{B}} = C$. Let $B \stackrel{\text{def}}{=} \frac{1}{\tau}\tilde{B}$. Hence, $e^{\tau B} = C$. Similarly, there exists a real matrix R such that $e^{2\tau R} = C^2$. Define

$$\begin{aligned}
P(t) &\stackrel{\text{def}}{=} \Phi(t)e^{-tB}, \\
Q(t) &\stackrel{\text{def}}{=} \Phi(t)e^{-tR}
\end{aligned}$$

so that $\Phi(t) = P(t)e^{tB} = Q(t)e^{tR}$ and that

$$P(t+\tau) = \Phi(t+\tau)e^{-\tau B}e^{-tB} = \Phi(t)CC^{-1}e^{-tB} = P(t),$$
$$Q(t+2\tau) = \Phi(t+2\tau)e^{-2\tau R}e^{-tR} = \Phi(t)C^2(C^2)^{-1}e^{-tR} = Q(t). \qquad \square$$

Remark 11.3.4. Observe that while Theorem 11.3.2 guarantees the existence of a matrix B such that $\Phi(\tau) = e^{B\tau}$, it makes no claims about uniqueness. This lack of uniqueness extends to the level of eigenvalues. If $\{\lambda_1, \ldots, \lambda_L\}$ are the eigenvalues of B then

$$\sigma_l = e^{\tau\lambda_l} = e^{\tau(\lambda_l + \frac{k}{\tau}2\pi i)}$$

where the last equality holds for all $k \in \mathbb{Z}$. What is unique is a real representation of the magnitude of the Floquet multipliers.

Definition 11.3.5. The *Lyapunov exponent* associated to the Floquet multiplier σ_l is defined by

$$\lambda_l := \frac{1}{\tau} \ln |\sigma_l| \in \mathbb{R}.$$

Note that using the notion of Lyapunov exponents, the stability of a periodic orbit can also be obtained, similarly as in Proposition 11.2.11.

Proposition 11.3.6. *Let $\Gamma = \{\gamma(t), t \in [0,\tau]\}$ be a τ-periodic orbit of the system* (11.1)*, $\{\sigma_l\}_{l=1,\ldots,L-1}$ be the nontrivial Floquet multipliers and $\{\lambda_l\}_{l=1,\ldots,L-1}$ be the corresponding set of Lyapunov exponents. Then*

(i) Γ is asymptotically stable if for all $l \in \{1, \ldots, L-1\}$, $\lambda_l < 0$;

(ii) Γ is unstable if there exists $l \in \{1, \ldots, L-1\}$ such that $\lambda_l > 0$.

Given a real $L \times L$ diagonalizable matrix A, let $\Sigma(A) = \{\alpha_l, v_l\}_{l=1,\ldots,L}$ denote the eigendecomposition of the square matrix A, i.e. $Av_l = \alpha_l v_l$, for all $l = 1, \ldots, L$. The following result shows how the information from the couple $(R, Q(t))$ coming from the real Floquet normal form $\Phi(t) = Q(t)e^{Rt}$ can directly be used to study the dynamical properties of the periodic orbit Γ, that is the stability of Γ can be determined by the eigenvalues of R while the stable and unstable normal bundles of Γ can be retrieved from the action of $Q(t)$ (with $t \in [0,\tau]$) on the eigenvectors of R.

Theorem 11.3.7. *Assume that $\Gamma = \{\gamma(t), t \in [0,\tau]\}$ is a τ-periodic orbit of* (11.1) *and consider $\Phi(t)$ the fundamental matrix solution of the non-autonomous linear equation $\dot{u} = Df(\gamma(t))u$ such that $\Phi(0) = I$. Suppose that a real Floquet normal form decomposition $\Phi(t) = Q(t)e^{Rt}$ is known. Assume that the real $n \times n$ matrix R is diagonalizable and let $\Sigma(R) = \{\mu_j, v_j\}_{j=1,\ldots,n}$ the eigendecomposition of R. Then the Lyapunov exponents l_j of Γ are given by*

$$l_j = Re(\mu_j). \tag{11.12}$$

Furthermore, for any $\theta \in [0, \tau]$, if one defines

$$w_j^\theta \stackrel{\text{def}}{=} Q(\theta)v_j, \tag{11.13}$$

then w_j^θ is an eigenvector of $\Phi_\theta(\tau)$ associated to the Lyapunov exponent l_j. Note that w_j^θ is a smooth 2τ-periodic function of θ.

Proof. Consider the eigendecomposition $\Sigma(R) = \{\mu_j, v_j\}_{j=1,\ldots,n}$ of the diagonalizable matrix R, meaning that the set $\{v_1, \ldots, v_n\}$ consists of n linearly independent eigenvectors of R. By Proposition 11.2.4, $\Phi(\tau)^2 = \Phi(2\tau)$. Since $Q(t)$ is 2τ-periodic and $Q(0) = I$, it follows that $\Phi(2\tau) = e^{R2\tau}$. Since R is diagonalizable, $\Phi(2\tau) = e^{R2\tau}$ is also diagonalizable. Since $\Phi(2\tau) = \Phi(\tau)^2$ and since the matrix $\Phi(\tau)$ is invertible and defined over the field of complex number (which has zero characteristic), then it can then be showed that $\Phi(\tau)$ is also diagonalizable. Now, since $\Phi(2\tau) = \Phi(\tau)^2$ one has that if $(\sigma, w) \in \Sigma(\Phi(\tau))$, then $(\sigma^2, w) \in \Sigma(\Phi(2\tau))$. Combining this last point with $\Phi(\tau)$, $\Phi(2\tau)$ being diagonalizable implies that the eigenspaces of $\Phi(\tau)$ and $\Phi(2\tau)$ are in one-to-one correspondence. That implies the existence of a set $\{\sigma_j\}_{j=1,\ldots,n}$ such that $\Sigma(\Phi(\tau)) = \{\sigma_j, v_j\}_{j=1,\ldots,n}$. From the property of the exponential matrix operator, $\Sigma(\Phi(2\tau)) = \{e^{\mu_j 2\tau}, v_j\}_{j=1,\ldots,n} = \Sigma(\Phi(\tau)^2) = \{\sigma_j^2, v_j\}_{j=1,\ldots,n}$. This implies that $\sigma_j^2 = e^{\mu_j 2\tau}$ for any $j = 1, \ldots, n$. Note that $l_j = Re(\mu_j)$ is the unique real number so that $|\sigma_j| = e^{l_j \tau}$. Hence, l_j is a Lyapunov exponent associated to the Floquet multipliers σ_j.

Now, from (11.7), one has that

$$\Phi_\theta(2\tau) = \Phi(2\tau + \theta)\Phi(\theta)^{-1} = Q(\theta)e^{(2\tau+\theta)R}e^{-R\theta}Q(\theta)^{-1}, \quad \forall \theta \in [0, \tau]$$

thus

$$\Phi_\theta(2\tau)Q(\theta)v_j = Q(\theta)e^{2\tau R}v_j = e^{2\tau\mu_j}Q(\theta)v_j$$

showing that $\Sigma(\Phi_\theta(2\tau)) = \{e^{2\tau\mu_j}, Q(\theta)v_j\}$. Applying the same argument than above, one can conclude that $\Sigma(\Phi_\theta(\tau)) = \{\sigma_j, Q(\theta)v_j\}_{j=1,\ldots,n}$ forms an eigendecomposition of the matrix $\Phi_\theta(\tau)$. Hence, $w_j^\theta = Q(\theta)v_j$ is an eigenvector of $\Phi_\theta(\tau)$. By the smoothness and the 2τ-periodicity of the matrix function $Q(\theta)$, one can conclude that $w_j^\theta = Q(\theta)v_j$ is also a smooth 2τ-periodic function of θ. □

Definition 11.3.8. Assume Γ is a hyperbolic periodic orbit. Let $w_j^\theta = a_j^\theta + ib_j^\theta$ be defined by (11.13). The *stable* and *unstable subspaces*, E_s^θ and E_u^θ, respectively, of the periodic orbit Γ at the point $\gamma(\theta)$ are defined by

$$E_s^\theta := \text{Span}\left\{a_j^\theta, b_j^\theta : |\sigma_j| < 1\right\}$$
$$E_u^\theta := \text{Span}\left\{a_j^\theta, b_j^\theta : |\sigma_j| > 1\right\}.$$

Definition 11.3.9. The *stable* and *unstable normal bundles* of Γ, $E^s(\Gamma)$ and $E^u(\Gamma)$, respectively, are defined by

$$E^s(\Gamma) := \bigcup_{\theta\in[0,\tau]} \{\gamma(\theta)\} \times E_s^\theta \quad \text{and} \quad E^u(\Gamma) := \bigcup_{\theta\in[0,\tau]} \{\gamma(\theta)\} \times E_u^\theta. \tag{11.14}$$

11.4 Computing the normal bundles and examples

From Definition 11.3.8, it is clear that constructing the stable and unstable normal bundles defined in (11.14) requires computing the 2τ-periodic function w_j^θ given in (11.13). We present two approaches to do so.

The first one, while more computationally expensive, leads to the direct computation of the couple $(R, Q(t))$ defining the real Floquet normal form. This is presented in Section 11.4.1. The second approach bypasses the computation of the normal form and instead provides a more direct way to obtain the function w_j^θ by solving an eigenvalue problem. This technique is introduced in Section 11.4.2

11.4.1 Computing the real Floquet normal form

Assume that $\gamma(t)$ is a τ-periodic solution of $\dot{x} = f(x)$ and denote $A(t) \stackrel{\text{def}}{=} Df(\gamma(t))$.

Recall that the computation of the real Floquet normal form (11.11) of the principal fundamental matrix Φ of $\dot{u} = A(t)u$ requires computing the couple $(R, Q(t))$, where $R \in M_n(\mathbb{R})$ and $Q : \mathbb{R} \to M_n(\mathbb{R})$ is a 2τ-periodic matrix valued function.

Differentiating the normal form $\Phi(t) = Q(t)e^{Rt}$ leads to

$$A(t)Q(t)e^{Rt} = A(t)\Phi(t) = \dot{\Phi}(t) = \dot{Q}(t)e^{Rt} + Q(t)Re^{Rt}.$$

Multiplying the above equation by e^{-Rt} and noticing that $I = \Phi(0) = Q(0)e^{R\cdot 0} = Q(0)$ lead to the problem

$$\begin{cases} \dot{Q}(t) + Q(t)R = A(t)Q(t) \\ Q(0) = I \end{cases} \tag{11.15}$$

The assumption on $Q(t)$ to be real and 2τ-periodic allows to consider the expansion

$$Q(t) = \sum_{k\in\mathbb{Z}} \mathcal{Q}_k e^{ik\frac{2\pi}{2\tau}t}, \tag{11.16}$$

where the Fourier coefficients $\mathcal{Q}_k \in M_n(\mathbb{C})$ for any $k \in \mathbb{Z}$. Being τ-periodic, the matrix-valued function $A(t) = Df(\gamma(t))$ is also 2τ-periodic, thus it makes sense to consider the expansion

$$A(t) = \sum_{k\in\mathbb{Z}} \mathcal{A}_k e^{ik\frac{2\pi}{2\tau}t}, \tag{11.17}$$

where $\mathcal{A}_k \in M_n(\mathbb{C})$ for any $k \in \mathbb{Z}$. Since $A(t)$ is τ-periodic, then $\mathcal{A}_k = 0$ for k odd and $\mathcal{A}_{2l} = \hat{\mathcal{A}}_l$ where $\hat{\mathcal{A}}_l$ is the l-th Fourier coefficient of $A(t)$ in the basis $\{e^{ik\frac{2\pi}{\tau}t}\}_k$.

After substituting the expansions (11.16) and (11.17) in problem (11.15), the latter system of ODEs moves into an equation $F(t) = 0$, where $F(t)$ is a 2τ-periodic matrix function. By a subsequent projection of $F(t)$ in the Fourier basis $\{e^{ik\frac{2\pi}{2\tau}t}\}$, it follows that solving (11.15) is equivalent to solve for the unknowns

$$R \in M_n(\mathbb{R}) \ \text{ and } \ \mathcal{Q} = (\mathcal{Q}_k)_{k\in\mathbb{Z}}, \quad \mathcal{Q}_k \in M_n(\mathbb{C})$$

the infinite dimensional algebraic system

$$F(x) = \begin{pmatrix} \eta(\mathcal{Q}) \\ f(R,\mathcal{Q}) \end{pmatrix} = 0 \tag{11.18}$$

where $x \stackrel{\text{def}}{=} (R, \mathcal{Q})$, and where η and $f = (f_k)_{k\in\mathbb{Z}}$ are given by

$$\eta(\mathcal{Q}) \stackrel{\text{def}}{=} \sum_{k\in\mathbb{Z}} \mathcal{Q}_k - I$$
$$f_k(R,\mathcal{Q}) \stackrel{\text{def}}{=} ik\frac{2\pi}{2\tau}\mathcal{Q}_k + \mathcal{Q}_k R - (\mathcal{A} * Q)_k$$

where $\mathcal{A} = (\mathcal{A}_k)_{k\in\mathbb{Z}}$ and $(\mathcal{A} * \mathcal{Q})_k$ denotes the *matrix convolution*

$$(\mathcal{A} * \mathcal{Q})_k \stackrel{\text{def}}{=} \sum_{k_1+k_2=k} \mathcal{A}_{k_1}\mathcal{Q}_{k_2}.$$

Using the radii polynomial approach (similar to the one presented in Chapter 10), one can find $\tilde{x} = (\tilde{R}, \widetilde{\mathcal{Q}})$ such that $F(\tilde{x}) = 0$. Having the exact solution, together with rigorous error bounds, allows computing the eigendecomposition of $\tilde{R}$ (using the theory of Section 4.5), which we denote by $\Sigma(\tilde{R}) = \{\mu_j, v_j\}_{j=1,\dots,n}$. Assume that the last eigenpair (μ_n, v_n) corresponds to the trivial Floquet multiplier. Using (11.16) and recalling (11.13), we set, for each $j = 1, \dots, n-1$,

$$w_j^\theta \stackrel{\text{def}}{=} Q(\theta)v_j = \sum_{k\in\mathbb{Z}} \widetilde{\mathcal{Q}}_k v_j e^{ik\frac{2\pi}{2\tau}\theta}.$$

11.4.2 Solving the eigenvalue problems

While the approach of Section 11.4.1 is appealing in the sense that it provides directly the real Floquet normal form, it leads to a formulation of a zero finding problem in (11.18) which involves many variables. More precisely, this approach requires solving for the Fourier coefficients of each the n^2 component of $Q(t)$, which can be computationally expensive, as the dimension n of the vector grows. In this section, we propose an alternative, more direct, approach to obtain the function w_j^θ given by (11.13).

This is done first by differentiating w_j^θ with respect to θ which, when combined with the first equation of (11.15) and with the fact that $Rv_j = \mu_j v_j$, lead to

$$\begin{aligned}\frac{d}{d\theta}w_j^\theta &= \frac{d}{d\theta}Q(\theta)v_j \\ &= \dot{Q}(\theta)v_j \\ &= (A(\theta)Q(\theta) - Q(\theta)R)v_j \\ &= A(\theta)(Q(\theta)v_j) - Q(\theta)(Rv_j) \\ &= Df(\gamma(\theta))w_j^\theta - \mu_j w_j^\theta.\end{aligned}$$

Hence, this leads (dropping the superscript θ and the index j) to the *eigenvalue problem*

$$\dot{w} + \mu w = Df(\gamma)w, \tag{11.19}$$

where $w : \mathbb{R} \to \mathbb{R}^n$ is a 2τ-periodic function (the eigenfunction) and $\mu \in \mathbb{C}$ is the eigenvalue with corresponding Lyapunov exponent $l \stackrel{\text{def}}{=} Re(\mu)$.

If there are n_s stable Floquet multiplier σ_j with $|\sigma_j| < 1$, then only n_s eigenvalue problems of the form (11.19) need to be solved to obtain n_s *eigenpairs* $(\mu_j, w_j(\theta))$ $(j = 1, \ldots, n_s)$ and hence to define the stable normal bundle $E^s(\Gamma)$ of Γ, as defined in Definition 11.3.9.

To compute an eigenpair $(\mu, w(t))$ of (11.19), we first proceed as in Section 11.4.1, that is as in (11.17), consider the expansion

$$A(t) = Df(\gamma(t)) = \sum_{k\in\mathbb{Z}} \mathcal{A}_k e^{ik\frac{2\pi}{2\tau}t}.$$

The assumption on $w(t)$ to be real and 2τ-periodic allows to consider the expansion

$$w(t) = \sum_{k\in\mathbb{Z}} w_k e^{ik\frac{2\pi}{2\tau}t}, \tag{11.20}$$

where the Fourier coefficients $w_k \in \mathbb{C}^n$ for any $k \in \mathbb{Z}$. Plugging the (2τ-periodic) expansions of $Df(\gamma(t))$ and $w(t)$ in (11.19) leads to the algebraic system $f = (f_k)_{k\in\mathbb{Z}} = 0$, where

$$f_k(\mu, w) \stackrel{\text{def}}{=} \left(ik\tfrac{2\pi}{2\tau} + \mu\right) w_k - (\mathcal{A} * w)_k \in \mathbb{C}^n$$

and where

$$(\mathcal{A} * w)_k = \sum_{k_1+k_2=k} \mathcal{A}_{k_1} w_{k_2} \in \mathbb{C}^n.$$

Imposing a phase condition on the eigenvector w (e.g. we can set the value of $w(0)$ to be equal to a fixed value), leads to a condition $\eta(w) = 0$. Combining the above leads to the zero finding problem

$$F(x) = \begin{pmatrix} \eta(w) \\ f(\mu, w) \end{pmatrix} = 0 \tag{11.21}$$

where $x \stackrel{\text{def}}{=} (\mu, w)$, which can be solved using the radii polynomial approach.

11.4.3 Examples

We now present applications of the proposed approach to compute stable and unstable normal bundles in the Lorenz equations and in the so-called ζ^3-model.

Example 11.4.1. Consider once again the Lorenz equations with the choice $\sigma = 10$ and $\beta = 8/3$. It is known that there exists a branch of periodic solutions parametrized by ρ joining a Hopf bifurcation at $\rho = \frac{470}{19} \approx 24.736$ and a homoclinic orbit at $\rho \approx 13.9265$. Figure 11.1(a-b) shows a bifurcation diagram and some of the periodic orbits of the continuous family. The computation of the periodic orbits and their normal stable and

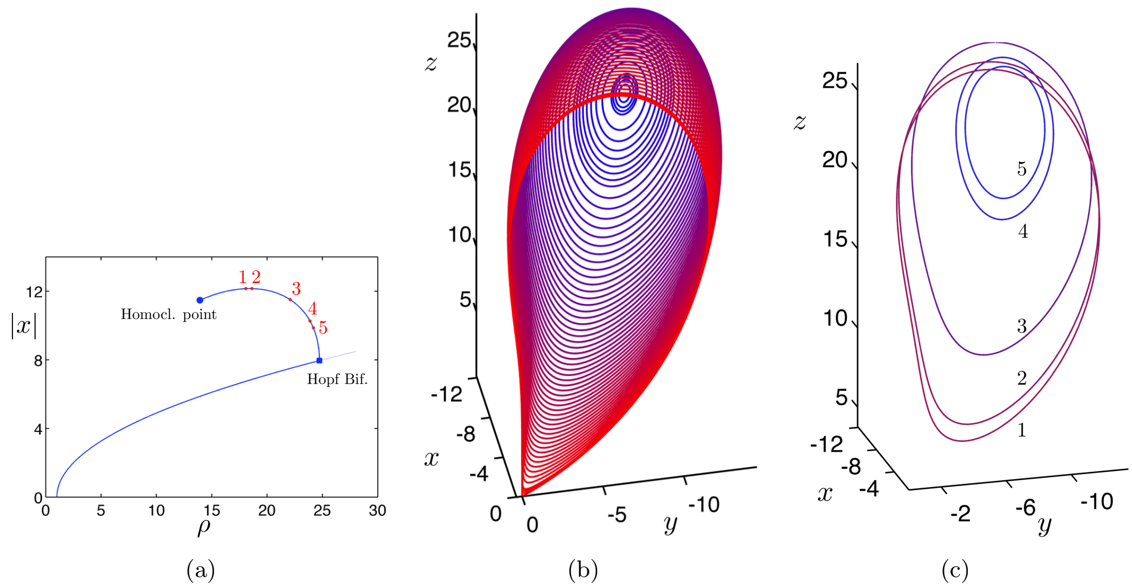

Figure 11.1: (a) A bifurcation diagram for the Lorenz system. The labelled points correspond to the values ρ_i. (b) Some of the periodic orbits on the family joining the Hopf Bifurcation and the homoclinic point. (c) The periodic solutions corresponding to $\rho = \rho_i$.

unstable bundles have been performed for a set of different periodic orbits of the Lorenz system lying on the mentioned bifurcation branch and corresponding to values of ρ, namely $\rho_1 = 18.0815$, $\rho_2 = 18.6815$, $\rho_3 = 20.8815$, $\rho_4 = 23.8815$ and $\rho_5 = 24.1816$. Figure 11.1(c) reports the graphics of the numerical approximation $\bar{\gamma}_i$ of the orbits corresponding to the choice ρ_i.

The first 15 Fourier coefficients of $\bar{\gamma}_1$ and $\bar{\gamma}_4$ are displayed in Table 11.1 and Table 11.2. The period of $\bar{\gamma}_1$ is approximately $\bar{\tau} = 1.027854840752128$ while the period of $\bar{\gamma}_4$ is about $\bar{\tau} = 0.683813590045746$.

Table 11.3 lists the Lyapunov exponents of the periodic orbits while in Figure 11.2 the tangent bundles are depicted.

$$\bar{a}_0 = \begin{pmatrix} -4.606354666884038 \\ -4.606354666884038 \\ 13.533127936581090 \end{pmatrix}$$

$\bar{a}_1$	$\bar{a}_2$	$\bar{a}_3$
$-2.457589444025310 - 0.734232230617879i$	$-0.840331595816763 - 0.280556868231764i$	$-0.244051806246999 - 0.079599377946577i$
$-2.008759805775985 - 2.236534819812307i$	$-0.497327754727251 - 1.307931343042055i$	$-0.098076628971404 - 0.527159479549944i$
$3.644641521462053 - 2.592383847330301i$	$1.373048803380924 - 0.590924707028489i$	$0.516674965708581 - 0.103578564294281i$

$\bar{a}_4$	$\bar{a}_5$	$\bar{a}_6$
$-0.066251105459624 - 0.023092599715315i$	$-0.017586386797219 - 0.007027398326009i$	$-0.004633846567037 - 0.002168050944621i$
$-0.009785901431526 - 0.185087447977630i$	$0.003892543961801 - 0.060779408215756i$	$0.003318015096906 - 0.019163826327315i$
$0.181193087615871 - 0.007305965402112i$	$0.059967166034911 + 0.005022757182874i$	$0.019005700003600 + 0.003642138482031i$

$\bar{a}_7$	$\bar{a}_8$	$\bar{a}_9$
$-0.001215481367863 - 0.000664841070850i$	$-0.000317263424301 - 0.000201467445482i$	$-0.000082326458994 - 0.000060305586234i$
$0.001629398704351 - 0.005865931787398i$	$0.000667978649296 - 0.001752989942270i$	$0.000249451961957 - 0.000513234481766i$
$0.005830102115005 + 0.001711097840891i$	$0.001743196289261 + 0.000687999808866i$	$0.000510298663577 + 0.000254394262682i$

$\bar{a}_{10}$	$\bar{a}_{11}$	$\bar{a}_{12}$
$-0.000021217174006 - 0.000017849897868i$	$-0.000005425393706 - 0.000005230883352i$	$-0.000001374889998 - 0.000001519316512i$
$0.000087897665531 - 0.000147548597224i$	$0.000029748123926 - 0.000041712327941i$	$0.000009770046204 - 0.000011604812976i$
$0.000146654700161 + 0.000089133299138i$	$0.000041443384209 + 0.000030059169428i$	$0.000011525197547 + 0.000009848310635i$

$\bar{a}_{13}$	$\bar{a}_{14}$	$\bar{a}_{15}$
$-0.000000344798445 - 0.000000437771341i$	$-0.000000085406182 - 0.000000125225422i$	$-0.000000020840071 - 0.000000035583174i$
$0.000003134076053 - 0.000003177810272i$	$0.000000986282445 - 0.000000856137980i$	$0.000000305435097 - 0.000000226673423i$
$0.000003154570627 + 0.000003153649381i$	$0.000000849430424 + 0.000000991127939i$	$0.000000224755146 + 0.000000306618231i$

Table 11.1: Fourier coefficients of the periodic orbit γ_1.

$$\bar{a}_0 = \begin{pmatrix} -7.521252250993276 \\ -7.521252250993276 \\ 22.399077327399255 \end{pmatrix}$$

$\bar{a}_1$	$\bar{a}_2$	$\bar{a}_3$
$-1.246453092091490 - 1.262394967959499i$	$-0.114587819240451 - 0.194828460289262i$	$-0.008589210614442 - 0.020708730881492i$
$-0.117098081743200 + 1.579694736267260i$	$0.015748373138453 + 0.188342422139680i$	$0.002967722490543 + 0.020459531930793i$
$-1.138858133842916 - 1.528289275687748i$	$-0.143528439361466 - 0.149121347835789i$	$-0.016769839350022 - 0.016159568619679i$

$\bar{a}_4$	$\bar{a}_5$	$\bar{a}_6$
$-0.000619073947482 - 0.001950094567689i$	$-0.000044052745410 - 0.000171023517657i$	$-0.000003073584415 - 0.000014085485779i$
$0.000325236831105 + 0.001967969477674i$	$0.000031364652698 + 0.000173341352082i$	$0.000002859395743 + 0.000014272552631i$
$-0.001814442573257 - 0.001764257857272i$	$-0.000188148978551 - 0.000185385380347i$	$-0.000018837629098 - 0.000018676999462i$

$\bar{a}_7$	$\bar{a}_8$	$\bar{a}_9$
$-0.000000208407126 - 0.000001089211660i$	$-0.000000013587818 - 0.000000078416175i$	$-0.000000000836854 - 0.000000005128148i$
$0.000000251244910 + 0.000001104010427i$	$0.000000021464593 + 0.000000079624828i$	$0.000000001792309 + 0.000000005227527i$
$-0.000001824392597 - 0.000001813988748i$	$-0.000000171368842 - 0.000000170686207i$	$-0.000000015658534 - 0.000000015616179i$

$\bar{a}_{10}$	$\bar{a}_{11}$	$\bar{a}_{12}$
$-0.000000000046957 - 0.000000000284715i$	$-0.000000000002182 - 0.000000000010252i$	$-0.000000000000052 + 0.000000000000357i$
$0.000000000146750 + 0.000000000292804i$	$0.000000000011806 + 0.000000000010900i$	$0.000000000000935 - 0.000000000000305i$
$-0.000000001395354 - 0.000000001392990i$	$-0.000000000121501 - 0.000000000121396i$	$-0.000000000010352 - 0.000000000010351i$

$\bar{a}_{13}$	$\bar{a}_{14}$	$\bar{a}_{15}$
$0.000000000000006 + 0.000000000000135i$	$0.000000000000002 + 0.000000000000021i$	$0.000000000000000 + 0.000000000000003i$
$0.000000000000073 - 0.000000000000131i$	$0.000000000000006 - 0.000000000000020i$	$0.000000000000000 - 0.000000000000003i$
$-0.000000000000863 - 0.000000000000865i$	$-0.000000000000070 - 0.000000000000071i$	$-0.000000000000005 - 0.000000000000006i$

Table 11.2: Fourier coefficients of the periodic orbit γ_4.

Example 11.4.2 (Non-orientable stable and unstable normal bundles). It is known that if a Floquet multipliers of a periodic orbit is negative, then the corresponding tangent bundle is non-orientable. Moreover, in the case of a saddle periodic orbit of a three-dimensional system, the two non-trivial Floquet multipliers are real and their product is positive. Therefore both the tangent bundles are either orientable or non-orientable and, in the latter case, they are topologically equivalent to a Möbius strip, see [29].

ρ	Stable Lyapunov exponent	Unstable Lyapunov exponent
ρ_1	-14.2953855130260	0.6287188463595
ρ_2	-14.2174898849454	0.5508232182790
ρ_3	-13.9620493680589	0.2953827013923
ρ_4	-13.7210150091049	0.0543483424385
ρ_5	-13.7013292393391	0.0346625726730

Table 11.3: Lyapunov exponents for each of the periodic orbit γ_i.

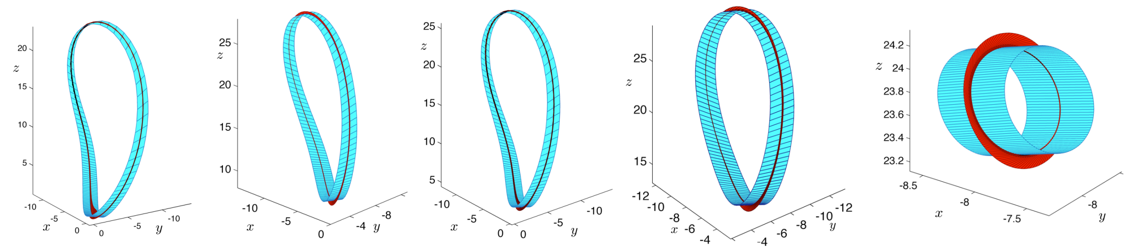

Figure 11.2: Plot of the tangent stable (turquoise) and unstable (red) bundles of each of the periodic orbits. From left to right: γ_1, γ_2, γ_3, γ_4 and γ_5.

An example of a dynamical system with periodic orbits that exhibit this behavior is the so called ζ^3-model considered in [2]

$$\begin{cases} \dot{x} = y \\ \dot{y} = z \\ \dot{z} = \alpha x - x^2 - \beta y - z. \end{cases} \tag{11.22}$$

For $\beta = 2$, as α varies, the periodic orbits of system (11.22) produce an interesting bifurcation diagram. We refer to [29] and [20] for a detailed analysis of the bifurcation diagram and on the genesis of periodic orbits, called twisted periodic orbit, with non orientable invariant manifolds. We focus on a particular twisted periodic orbit corresponding to $\alpha = 3.372$ lying on the branch emanating from a period-doubling bifurcation that occurs at $\alpha \approx 3.125$.

Following the same procedure as before, we compute the periodic orbit $\gamma(t)$ and we extract the necessary stability parameters and we recover the stable and unstable tangent bundles. Figure 11.3 shows the resulting bundles.

Having computed the period τ of the orbit, we realize that the absolute values of the two nontrivial Floquet multipliers satisfy

$$|\sigma_1| \approx 7.037235782193 \cdot 10^{-3} \quad \text{and} \quad |\sigma_2| \approx 1.526609276443494.$$

To conclude this example, we emphasize the role played by the continuous function $Q(\theta)$ in the construction of the tangent bundles. As proved in Theorem 11.3.7, as θ changes, the eigenvector w_j^θ of $\Phi_\theta(\tau)$ associated to the Floquet multiplier σ_j is given by $w_j^\theta = Q(\theta)v_j$, where v_j is the eigenvector of R relative to the eigenvalue μ_j. The function $Q(\theta)$ is continuous and 2τ-periodic, but the tangent bundles are smooth manifolds. That implies that $w_j^\tau = Q(\tau)v_j$ has to be an eigenvector of $\Phi(\tau)$ associated to the Floquet multiplier σ_j, i.e. $span\{v_j\} = span\{w_j^\tau\}$. In the case of the Lorenz system in Example 11.4.1 $Q(\tau)$ turns to be the identity matrix, therefore the last relation is simply verified. But in case of the ζ^3-model and in general when the bundles are not orientable $Q(\tau)$ need not be the identity matrix.

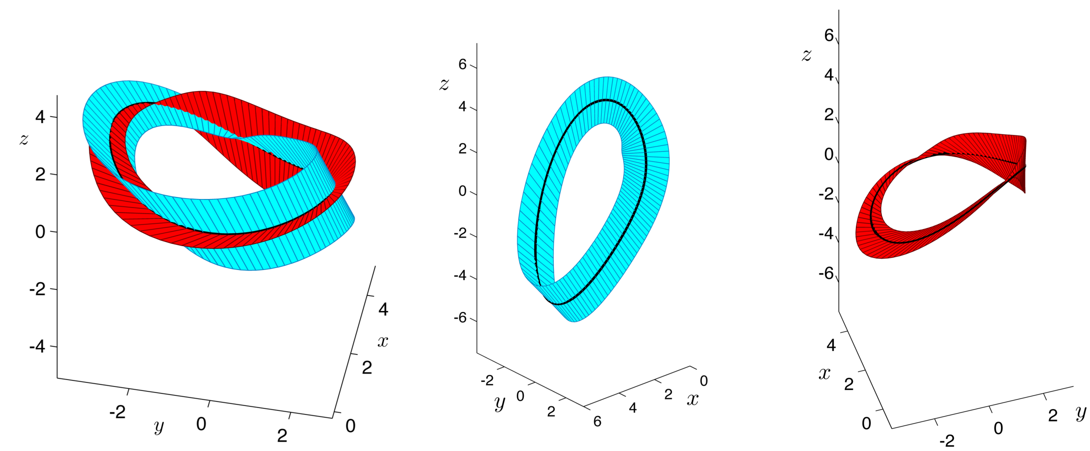

Figure 11.3: Stable (turquoise) and unstable (red) tangent bundles of a periodic orbit for the ζ^3 model with negative Floquet multipliers.

11.5 Stable and unstable manifolds of periodic orbits

Recall from Section 4.4 that stable and unstable manifolds to equilibria provide us with an understanding of how solutions asymptotically approach or leave the equilibria. We are now in a position to develop the analogous structures for periodic orbits.

Definition 11.5.1. Let Γ be a τ-periodic solution to (11.1). Let U be a neighborhood of Γ. The *local stable manifold* $W^s(\Gamma, U)$ and *local unstable manifold* $W^u(\Gamma, U)$ are defined as follows:

$$W^s(\Gamma, U) := \{x \in U \mid \varphi([0,\infty), x) \subset U \text{ and } \omega(x) = \Gamma\}$$

and

$$W^u(\Gamma, U) := \{x \in U \mid \varphi((-\infty, 0], x) \subset U \text{ and } \alpha(x) = \Gamma\}.$$

The *stable* and *unstable manifolds* are obtained by setting $U = \mathbb{R}^n$, i.e.

$$W^s(\Gamma) := \{x \in \mathbb{R}^n \mid \omega(x) = \Gamma\}$$

and

$$W^u(\Gamma) := \{x \in \mathbb{R}^n \mid \alpha(x) = \Gamma\}\,.$$

As stated above the language is a bit misleading. As in the case of fixed points, to guarantee that $W^s(\Gamma, U)$ and $W^u(\Gamma, U)$ are actually manifolds requires the assumption that Γ is a hyperbolic periodic orbit.

Theorem 11.5.2 (Stable and Unstable Manifold Theorem for Periodic Orbits). *Consider a smooth vector field $\dot{x} = f(x)$, $x \in \mathbb{R}^n$. Let φ denote the associated flow and let $\Gamma = \{\gamma(t) \mid t \in [0, \tau]\}$ be a τ-periodic orbit. Assume n_s of the Floquet multipliers have magnitude less than one and let $n_u \stackrel{\text{def}}{=} n - 1 - n_s$. Then, there exits an neighborhood U of Γ such that $W^s(\Gamma, U)$ is a n_s-dimensional, differentiable manifold which is positively invariant under the flow, and $W^u(\Gamma, U)$ is an n_u-dimensional, differentiable manifold which is negatively invariant under the flow. Furthermore, at Γ, $W^s(\Gamma, U)$ is tangent to the stable normal bundle $E^s(\Gamma)$ while $W^u(\Gamma, U)$ is tangent to the unstable normal bundle $E^u(\Gamma)$.*

Chapter 12

Parameterization Method for Stable and Unstable Manifolds for Periodic Orbits

12.1 Fourier-Taylor series expansions

Recall that the complex open strip of width $\rho > 0$ is the set

$$A_\rho = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < \rho\},$$

and the m-dimensional poly-disk of radius $r > 0$ about $w_0 \in \mathbb{C}^m$ is the set

$$D_r^m(w_0) = \left\{w = (w_1, \ldots, w_m) \in \mathbb{C}^m : |w_j - w_j^0| < r \text{ for all } 1 \leq j \leq m\right\},$$

Define the m-dimensional (ρ, r) poly-cylinder at w_0 to be the set

$$C_{\rho,r}^m(w_0) = A_\rho \times D_r^m(w_0).$$

A continuous function $F\colon C_{\rho,r}^m(w_0) \to \mathbb{C}$ is T-periodic in its first variable if

$$F(z + T, w) = F(z, w),$$

for all $z \in A_\rho$ and all $D_r^m(w_0)$, and is analytic if the complex partial derivatives

$$\frac{\partial}{\partial z} F(z, w), \qquad \text{and} \qquad \frac{\partial}{\partial w_j} F(z, w),$$

exist and are bounded for each $(z, w) \in C_{\rho,r}^m(w_0)$ and each $1 \leq j \leq m$, i.e. if F is analytic in each variable separately with the others held constants.

The appropriate notion of analyticity in this context is the existence of a convergent Fourier-Taylor series expansion at each point in $C_{\rho,r}^m(w_0)$. More precisely, and employing the multi-index notation of Chapter (Reference),